

TECHNOLOGY AT YOUR SERVICE

World's smallest hearing aid is almost invisible

A team of researchers has designed the world's smallest hearing aid, which is so tiny that it is almost invisible. Called Nanoplug, the hearing aid is meant to be less intrusive, less noticeable and less expensive than other such devices. It measures only $7.1 \times 5.7 \times 4.17\text{mm}$, which is half the size of any other hearing aid.

Nanoplug was proposed by Nevena Zivic, and later developed by industrial designer, Jongha Lee, audio engineer, Mladen Stavri and electromechanical engineer, Zoran Marinovi. The team claims that the hearing aid has no toxic chemicals or heavy metals inside. It can be programmed using software running on a computer, the results of which can be downloaded directly to the device via a cable. The device also reduces the risk of internal feedback and wind noise.

Autistic kids can now learn social and communication skills using Google Glass

Now, kids suffering from autism will be able to learn social and communication skills. A USA based firm is working on Google Glass apps and hardware add-ons, which will help them in learning these skills. Caregivers will be able to receive feedback from these apps too.

The start-up, Brain Power, based in Cambridge, Massachusetts, in collaboration with neuroscientist, Ned T. Sahin, founded the company.

Autism disorder is a common problem world-wide, and the company is focused on helping autistic children who are suffering from medium- to high-level autism. Brain Power has released the beta version and 200 people have already enrolled for it. The testing will be done in three phases.

Google Glass has found an ideal solution for kids suffering from autism because of its accelerometer chip that supports head gestures. This facility is used by Brain Power to track when kids are looking at their parents and when they are not. Tracking can also be done for stereotypical and repetitive movements, which are quite common among autism patients. Sahin said, the company's main motive is to offer numerical, as well as objective, assessment of behavioural patterns of a child.

Google Glass apps, designed by Brain Power, have been named Empowered Brain Suite for Autism. These apps will encourage kids to communicate better with their parents and also build eye-contact. Apps include social engagement module monitors, which will assess how a child engages with his/her parents. There is also a series of games that enable affected kids read/explain expressions. This suite



Google Glass

also addresses the issue of language. Google Glass portrays an object and its name gets displayed and pronounced through Google Glass ear-bud.

An app to help you land an airplane

Here comes an app that could help you land your plane in case the aircraft engine fails. The new iPad app, Xavion, can be connected to a small plane's autopilot via Wi-Fi to help guide the pilot to the nearest airport or safe-landing area. It extracts data from autopilot to provide additional aid to the pilot, who can use it to land the plane safely.

The app can help pilots navigate their way to safe landing spots, provide weight and balance checks, estimate fuel burn and time to the destination at various altitudes. A new update will let the app take control of the autopilot, making any human intervention unnecessary.

Boeing, BlackBerry to build self-destructing phones

BlackBerry is working with Boeing on building a high-security Android based smartphone called Boeing Black. Boeing Black can encrypt calls and messages. It can be very useful for government agencies that need to keep their data and communication secure. John Chen, CEO, BlackBerry, said that Boeing Black will be powered by BES 12 platform, which can also run Android apps. Boeing has collaborated with BlackBerry for the OS.

BES (BlackBerry Enterprise Service) v12 lets users manage and secure communication that is done with BlackBerry devices, as well as Android and iOS devices. This is an advanced level of security as compared to BlackBerry's traditional internal networks security.

Boeing has outsourced the hardware design to the aerospace and defence contractor in Chicago, USA. The hardware company from Chicago is known for its products for jet-liners and fighter planes. The company